Fiscal Year:	FY 2010	**Task Last Updated:**	FY 09/03/2010
PI Name:	Limoli, Charles Ph.D.		
Project Title:	Dose-rate and mixed field effects of protons and HZE nuclei on oxidative injury and stem cell plasticity in the CNS		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92697-2695	**Congressional District:**	45
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Space Radiation NNJ07ZSA001N
Start Date:	03/01/2008	**End Date:**	02/28/2010
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 2/28/2010 per A. Chu/ARC (3/2009)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX08AG10G		
Performance Goal No.:			
Performance Goal Text:			
	The radiation fields in space are unique and pose significant hazards to astronauts since severe biological damage can result when the energetic particles present in space pass through the cells and tissues of the body. The body has many sensitive cells, tissues and organ systems that can undergo significant abnormal changes when damaged by space radiation and many of these adverse effects can have long lasting consequences. The brain contains populations of radiation sensitive cells called neural stem and precursor cells. Under normal circumstances they divide and develop into new neurons throughout life, thereby contributing to the cognitive health of an individual. When the brain is exposed to the radiation in space these sensitive cells can die and/or accumulate changes that compromise their ability to help in the repair of radiation injury. Many of these radiation-induced changes are adverse in that they can lead to suboptimal cognitive performance, which for astronauts engaged in challenging tasks and situations, may be particularly problematic.		

Task Description:	One aspect this proposal will focus on is how radiation leads to the types of changes in the brain that might lead to impaired cognition. Much of our past work has demonstrated that the sensitive populations of neural stem and precursors cells that are important to cognitive health exhibit long-lasting chemical changes caused by prior radiation exposure. These changes have been linked to many of the adverse effects found in the brain after irradiation, and our present study will continue to identify the pathways that are activated by irradiation that cause these long-term and adverse chemical changes. Another aspect of our proposal relates to the rate at which cells are exposed to the radiation. A given dose of total radiation can be given over short or long time frames; if that exposure time is short then the rate is higher that if that dose is delivered over longer periods of time. Much of our past data of space radiation effects in the brain have been done under conditions of high dose rate, largely due to a number of practical limitations. In space however, most dose rates are low, and because the response of cells might differ, studies are needed to determine how cells respond to the types of radiation in space when delivered at low dose rates. Our current proposal will compare the types of changes found in neural precursor cells at low dose rate to those changes found previously at high dose rate. Results from these studies will be critical in enhancing our knowledge base of the risks associated with exposure of the brain to the radiation fields in space. In conclusion our research will analyze a population of radiation sensitive precursor cells in the brain that are critical to the cognitive health of an individual. These cells undergo long-lasting chemical changes when irradiated and these changes are linked to impaired cognitive performance. Our studies will employ neural precursor cells grown under a variety of conditions and that are manipulated at the genetic level, to understand how their response to low doses and dose rates of space radiation impact their function.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results from these studies will be critical in enhancing our knowledge base of the risks associated with exposure of the brain to the radiation fields in space.
Task Progress:	Data accumulated over this funding period has demonstrated that the radiation response of multipotent neural precursor cells to x-rays, protons, and HZE nuclei involves an acute and persistent oxidative stress that can last for months. The importance of neural precursor cells to the repair and remodeling of the irradiated CNS is underscored by our data showing that the radiation-induced depletion of these cells is linked to the inhibition of neurogenesis and the onset of cognitive impairment. We believe oxidative injury is critical in this context, and plays a contributory if not causal role in determining neural precursor cell function, survival and fate. We have shown that oxidative stress significantly alters proliferation, radiosensitivity, mitochondrial function, and differentiation, and have used these studies to advance our understanding of space radiation in the CNS. Significant new findings and accomplishments under this funding period are as follows: 1) We have optimized the growth conditions of our human neural stem cell system and have characterized these cells to low LET radiation modalities. We are now poised to subject these cells to charged particle irradiation. 2) We have generated new cell lines from mice deficient for the extracellular isoform of superoxide dismutase (EC-SOD). Studies have now been initiated to characterize how cells lacking this key antioxidant enzyme respond to ionizing radiation. 3) We have optimized the assessment of metabolic activity after low dose exposure using non-invasive 2-photon spectroscopy. 4) We have determined that dose rate along with total dose is an important parameter dictating the response of neural stem and precursor cells to proton irradiation. 5) We have expanded our prior studies showing that studies determining particle traversals at < or = 1 per cell elicit significant increases in oxidative stress. 6) We have also expanded our previous findings showing that low dose radiation-induced oxidative stress persists for many weeks to months following acute exposures. 7) We have significantly expanded our radioadaptive studies showing that prior exposure attenuates the increase in radiation-induced oxidative stress, leading to improved survival, enhanced DSB repair, and increased antioxidant capacity.
Bibliography Type:	Description: (Last Updated: 01/14/2026)